XRPL Validator — Daily Engineering Report · 2026-03-21

Period: 2026-03-21 00:00 to 24:00 US Eastern (2026-03-21 04:00 to 2026-03-22 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: git history only. Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data; data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 36 commits (10 with a recorded merge into main; xrpl-validator merge records begin 2026-07-04).

Service levels

No signing records for this day (they begin 2026-09-19).

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

Merged	Repo	Commit	Subject
15:51 UTC	xrpl-sdk	`4417b88`	fix: correct repository URL in all Cargo.toml files
15:53 UTC	xrpl-sdk	`b2d8b35`	docs: replace "production-grade" with "community-built"
16:05 UTC	xrpl-sdk	`3710bc3`	ci: add Dependabot for cargo and GitHub Actions updates
19:35 UTC	xrpl-sdk	`14b6e1b`	ci: bump actions/checkout from 4 to 6
19:35 UTC	xrpl-sdk	`753529d`	deps: bump tokio-tungstenite from 0.26.2 to 0.29.0
19:35 UTC	xrpl-sdk	`1ba8be6`	deps: bump rand from 0.8.5 to 0.9.2
19:35 UTC	xrpl-sdk	`fb7a327`	fix(core): update thread_rng() to rng() for rand 0.9 compatibility
19:35 UTC	xrpl-sdk	`a9f1a34`	Merge pull request #1 from ilubxrp589/dependabot/github_actions/actions/checkout-6
19:35 UTC	xrpl-sdk	`6117e6a`	Merge pull request #2 from ilubxrp589/dependabot/cargo/tokio-tungstenite-0.29.0
19:35 UTC	xrpl-sdk	`360662e`	Merge pull request #3 from ilubxrp589/dependabot/cargo/rand-0.9.2

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix: correct repository URL in all Cargo.toml files** (`4417b88` , xrpl-sdk) Update repository field from placeholder xrpl-rust org to actual ilubxrp589 account. Fixes docs.rs links, crates.io metadata, and cargo tree output.
- **fix(core): update thread_rng() to rng() for rand 0.9 compatibility** (`fb7a327` , xrpl-sdk) rand 0.9 renamed thread_rng() to rng().
- **feat: initial scaffold — xrpl-ledger and xrpl-node crates** (`4299395` , xrpl-validator) Two-crate workspace for building an XRPL validator node in Rust: - xrpl-ledger: SHAMap, NodeStore trait, ledger header/objects/state (module stubs + error types, depends on xrpl-core via git) - xrpl-node: peer protocol, consensus, mempool, RPC server (module stubs + config + error types, protobuf codegen from rippled's xrpl.proto, tracing framework)
- **feat(node): node identity, message types, and wire codec** (`07c32fd` , xrpl-validator)

- peer/identity.rs: NodeIdentity with Ed25519 keypair, self-signed TLS cert (pubkey in CN), rustls server/client configs, AcceptAnyCert verifier for self-signed peer network (4 tests) - peer/message.rs: PeerMessage enum wrapping all 21 protobuf message types with encode/decode dispatch by type code - peer/codec.rs: MessageCodec implementing tokio-util Decoder/Encoder with 4-byte length prefix, 2-byte type code, zlib …
- **feat(node): handshake, connection, peer lifecycle** (`d92281f` , xrpl-validator)
- peer/handshake.rs: outbound XRPL/2.0 handshake over TLS — HTTP upgrade with Public-Key, Session-Signature (TLS exporter keying material), Network-ID, Crawl headers. Supports mainnet/testnet/devnet. - peer/connection.rs: async read/write loops via tokio select!, auto ping/pong handling, overlay message forwarding, graceful shutdown via watch channel. PeerHandle for sending messages to peers. PeerInfo tracks pubkey, …
- **feat(node): complete M1 peer protocol — live testnet handshake working** (`ebbb373` , xrpl-validator) Major milestone: successfully handshake with XRPL testnet and decode peer protocol messages.
- **fix(node): M1 audit fixes — remaining bytes, docs, codec safety** (`140d13d` , xrpl-validator) Fixes from peer protocol audit: 1. CRITICAL: Pass remaining_bytes from handshake through to connection's codec buffer. The HTTP header read may consume bytes belonging to the binary protocol — these must be prepended to the Framed decoder's buffer.
- **fix(codec): correct frame length interpretation — messages now decode** (`12f9941` , xrpl-validator) The 4-byte length field in XRPL peer protocol frames is the protobuf payload size ONLY — it does NOT include the 2-byte type code.
- **feat(node): live XRPL peer message viewer on port 3777** (`e9613c4` , xrpl-validator) Web dashboard streaming real-time decoded XRPL peer protocol messages from the testnet via Server-Sent Events.
- **feat(viewer): add built-with credit line** (`f1cc9dc` , xrpl-validator)
- **fix(viewer): no-scroll page layout with contained feed box** (`2f3633d` , xrpl-validator) Page no longer scrolls — feed is in a bordered box that scrolls internally. Header and built-with text stay fixed.
- **feat(viewer): add (TESTNET) label to header** (`b7b30c0` , xrpl-validator)
- **feat(ledger): SHAMap implementation — hash prefixes, nodes, trie** (`04f795c` , xrpl-validator) M2 progress: core SHAMap data structure for XRPL ledger state.
- **feat(ledger): complete M2 — SHAMap, NodeStore, ledger state (audited)** (`1b424d4` , xrpl-validator) Full ledger storage implementation, audited twice with all findings fixed.
- **feat(viewer): Ripple-branded dashboard with live visualizations** (`8d2cc9b` , xrpl-validator) Complete UI overhaul with XRP/Ripple brand colors (#0085FF primary, dark navy backgrounds) and Orbitron + JetBrains Mono typography.
- **feat(viewer): peer connection list display** (`0dbac6e` , xrpl-validator) Backend sends peer endpoint data from TMEndpoints messages via SSE. Frontend displays peers as chips in a bar between the type distribution and stats row. Shows endpoint address and hop count. Updates each time the connected peer sends its peer list (~every 60s).
- **feat(viewer): live validator consensus grid** (`e3f904a` , xrpl-validator) Each validator is now tracked individually during consensus: - Proposals: extract node_pub_key from TMProposeSet, dot turns blue - Validations: extract SigningPubKey from serialized validation blob by scanning for field code 0x73 0x21, dot turns yellow - Consensus reached (ACCEPTED): all active dots flash green
- **feat(viewer): dedicated consensus monitor page at /consensus** (`a4ffd3c` , xrpl-validator) Full-screen validator agreement visualization: - Big ledger counter (Orbitron 48px, blue flash on close) - Phase indicator: PROPOSING

→ CLOSING → VALIDATING → CONSENSUS - Validator grid: 16x16px nodes that light up as they participate - Idle (dark) → Proposed (blue) → Validated (yellow) → Agreed (green) - Tooltip on hover shows validator key prefix - Stats: proposals, validations, unique validators seen, close …

- **feat(viewer): nebula background, physics orbs, rainbow consensus** (`853e04e`, xrpl-validator) Complete consensus page redesign: - Animated nebula background (5 drifting gradient blobs, 80px blur) - Force-directed validator orbs with orbital spin, gravity, repulsion - State effects: proposed (blue burst), validated (yellow shockwave), agreed (green explosion + outward blast + gravity pull-back) - Rainbow particle explosion on consensus: 80 HSL-cycling particles from center + 12 per agreed node, with glow …
- **feat(node): complete M3 transaction engine with live sig verification** (`604c1c1`, xrpl-validator) Full transaction validation pipeline with codec integration (audited): mempool/validate.rs: - Decode tx blob via xrpl-core binary codec - Extract Account, Fee, Sequence, LastLedgerSequence from decoded JSON - Signature verification: re-encode for signing, compute signing hash - Ed25519: verify against raw STX\0||bytes (Ed25519 does internal hash) - Secp256k1: verify against SHA512Half(STX\0||bytes) - Fee sanity …
- **feat(node): complete M4 consensus engine (audited)** (`bf78b0a`, xrpl-validator) Full RPCA consensus implementation: - unl.rs: Trusted validator list, manifest key rotation - state.rs: Open/Establish/Accept state machine with stability check - threshold.rs: Dynamic convergence 50%→60%→70%→80%→95% - close_time.rs: Resolution rounding, mode voting - validation.rs: Per-ledger validation tracking, >=80% fully-validated - proposal.rs: Transaction set hash (single source, re-exported) - …
- **feat: complete all 40 cards + metrics page + full audit fixes** (`ab2ad99`, xrpl-validator) Final milestone — all 40 taskboard cards implemented and audited.
- **feat: switch to XRPL mainnet + domain verification** (`48544b2`, xrpl-validator) val-041: Switched from testnet to mainnet - Peer: s1.ripple.com:51235 (was s.altnet.rippletest.net) - Network-ID: 0 (mainnet, was 1 testnet) - All 3 pages updated: MAINNET label
- **feat: sled storage, PM2 service, 3D consensus, GitHub link** (`eec04a2`, xrpl-validator) val-043: Sled persistent NodeStore behind 'sled-backend' feature val-054: PM2 'xrpl-validator' service on release binary
- **feat(node): fast parallel ledger sync — 1500+ objects/sec from mainnet** (`c93fef7`, xrpl-validator) val-044: sync_ledger binary downloads XRPL mainnet state objects.
- **feat(viewer): tx/s rate on feed + metrics pages** (`cb4a1ab`, xrpl-validator) Feed page: tx/s in stats row, updated every second Metrics page: Transactions/Second card with purple sparkline graph
- **feat(viewer): live ledger sync progress bar on feed + metrics** (`3dc97a2`, xrpl-validator) /api/sync-status endpoint reads sync file from NVMe drive, returns object count, estimated percentage, file size, and active status.
- **feat(viewer): multi-peer mainnet connections** (`59d512f`, xrpl-validator) Connect to s1.ripple.com AND s2.ripple.com simultaneously. Each peer runs its own connection loop with auto-reconnect. Both feed into the same SSE stream — double the network visibility.

Docs and operations

- **docs: replace "production-grade" with "community-built"** (`b2d8b35`, xrpl-sdk)
- **ci: add Dependabot for cargo and GitHub Actions updates** (`3710bc3`, xrpl-sdk)
- **ci: bump actions/checkout from 4 to 6** (`14b6e1b`, xrpl-sdk)
- **deps: bump tokio-tungstenite from 0.26.2 to 0.29.0** (`753529d`, xrpl-sdk)
- **deps: bump rand from 0.8.5 to 0.9.2** (`1ba8be6`, xrpl-sdk)

- **Merge pull request #1 from ilubxrp589/dependabot/github_actions/actions/checkout-6** (`a9f1a34` , xrpl-sdk)
- **Merge pull request #2 from ilubxrp589/dependabot/cargo/tokio-tungstenite-0.29.0** (`6117e6a` , xrpl-sdk)
- **Merge pull request #3 from ilubxrp589/dependabot/cargo/rand-0.9.2** (`360662e` , xrpl-sdk)

Independence program

No independence record for UTC day 2026-03-21; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
03-21 15:51	03-21 15:51	xrpl-sdk	`4417b88`	fix: correct repository URL in all Cargo.toml files
03-21 15:53	03-21 15:53	xrpl-sdk	`b2d8b35`	docs: replace "production-grade" with "community-built"
03-21 16:05	03-21 16:05	xrpl-sdk	`3710bc3`	ci: add Dependabot for cargo and GitHub Actions updates
03-21 16:06	03-21 19:35	xrpl-sdk	`14b6e1b`	ci: bump actions/checkout from 4 to 6
03-21 16:06	03-21 19:35	xrpl-sdk	`753529d`	deps: bump tokio-tungstenite from 0.26.2 to 0.29.0
03-21 16:06	03-21 19:35	xrpl-sdk	`1ba8be6`	deps: bump rand from 0.8.5 to 0.9.2
03-21 19:33	03-21 19:35	xrpl-sdk	`fb7a327`	fix(core): update thread_rng() to rng() for rand 0.9 compatibility
03-21 19:35	03-21 19:35	xrpl-sdk	`a9f1a34`	Merge pull request #1 from ilubxrp589/dependabot/github_actions/actions/checkout-6
03-21 19:35	03-21 19:35	xrpl-sdk	`6117e6a`	Merge pull request #2 from ilubxrp589/dependabot/cargo/tokio-tungstenite-0.29.0
03-21 19:35	03-21 19:35	xrpl-sdk	`360662e`	Merge pull request #3 from ilubxrp589/dependabot/cargo/rand-0.9.2
03-21 20:26	—	xrpl-validator	`4299395`	feat: initial scaffold — xrpl-ledger and xrpl-node crates

Committed	Merged into main	Repo	Commit	Subject
03-21 20:32	—	xrpl-validator	`07c32fd`	feat(node): node identity, message types, and wire codec
03-21 20:39	—	xrpl-validator	`d92281f`	feat(node): handshake, connection, peer lifecycle
03-21 20:59	—	xrpl-validator	`ebbb373`	feat(node): complete M1 peer protocol — live testnet handshake working
03-21 21:13	—	xrpl-validator	`140d13d`	fix(node): M1 audit fixes — remaining bytes, docs, codec safety
03-21 21:23	—	xrpl-validator	`12f9941`	fix(codec): correct frame length interpretation — messages now decode
03-21 21:28	—	xrpl-validator	`e9613c4`	feat(node): live XRPL peer message viewer on port 3777
03-21 21:31	—	xrpl-validator	`f1cc9dc`	feat(viewer): add built-with credit line
03-21 21:33	—	xrpl-validator	`2f3633d`	fix(viewer): no-scroll page layout with contained feed box
03-21 21:43	—	xrpl-validator	`b7b30c0`	feat(viewer): add (TESTNET) label to header
03-21 22:48	—	xrpl-validator	`04f795c`	feat(ledger): SHAMap implementation — hash prefixes, nodes, trie
03-21 23:10	—	xrpl-validator	`1b424d4`	feat(ledger): complete M2 — SHAMap, NodeStore, ledger state (audited)
03-21 23:16	—	xrpl-validator	`8d2cc9b`	feat(viewer): Ripple-branded dashboard with live visualizations
03-21 23:31	—	xrpl-validator	`0dbac6e`	feat(viewer): peer connection list display
03-21 23:48	—	xrpl-validator	`e3f904a`	feat(viewer): live validator consensus grid
03-21 23:52	—	xrpl-validator	`a4ffd3c`	feat(viewer): dedicated consensus monitor page at /consensus
03-22 00:41	—	xrpl-validator	`853e04e`	feat(viewer): nebula background, physics orbs, rainbow consensus
03-22 01:17	—	xrpl-validator	`604c1c1`	feat(node): complete M3 transaction engine with live sig verification
03-22 01:31	—	xrpl-validator	`bf78b0a`	feat(node): complete M4 consensus engine (audited)
03-22 02:00	—	xrpl-validator	`ab2ad99`	feat: complete all 40 cards + metrics page + full audit fixes
03-22 02:14	—	xrpl-validator	`48544b2`	feat: switch to XRPL mainnet + domain verification
03-22 02:47	—	xrpl-validator	`eec04a2`	feat: sled storage, PM2 service, 3D consensus, GitHub link
03-22 03:18	—	xrpl-validator	`c93fef7`	feat(node): fast parallel ledger sync — 1500+ objects/sec from mainnet
03-22 03:41	—	xrpl-validator	`cb4a1ab`	feat(viewer): tx/s rate on feed + metrics pages
03-22 03:48	—	xrpl-validator	`3dc97a2`	feat(viewer): live ledger sync progress bar on feed + metrics
03-22 03:59	—	xrpl-validator	`59d512f`	feat(viewer): multi-peer mainnet connections

Times in UTC. A commit appears on the day it was committed to main or merged into it.